

Genome analysis

Puzzler: scalable one-command platinum-quality genome assembly from HiFi and Hi-C

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Abstract

Motivation: Chromosome-level assemblies are essential for modern genomics, from comparative genomics and evolutionary studies to precision breeding. While integrated HiFi and Hi-C data now enable accurate chromosome-scale genome assemblies, the bioinformatic process remains complex and involves specialized tools and expertise. With large-scale pan-genomic efforts requiring dozens to hundreds of platinum quality chromosome-scale genomes, there is a need for scalable, portable, and user-friendly pipelines that streamline and standardize high-quality genome assembly workflows.

Results: We introduce Puzzler, a containerized, scalable pipeline for chromosome-scale *de novo* genome assembly using PacBio HiFi and Hi-C data. Designed for portability and minimal user input, Puzzler automates contig assembly, duplicate purging, Hi-C-based scaffolding, and chromosome assignment via synteny, even with highly diverged reference taxa. Optional modules generate input files for manual Hi-C curation or operate reference-free. Quality control is integrated and includes Hi-C contact maps, BUSCO, yak k-mer completeness, and BlobTools contamination screening. A checkpointing system ensures that previously completed tasks are not re-executed, while a simple sample sheet input structure supports scalable batch processing. Puzzler has been validated on genomes ranging from 24 Mbp to 6.5 Gbp, delivering highly contiguous assemblies with <10 min of user input, enabling high-throughput platinum-quality genome assembly.

Availability and implementation: Puzzler is released into the public domain under 17 U.S.C. §105. Source code, documentation, and tutorials are available at <https://github.com/merondun/puzzler> and archived on Zenodo: <https://doi.org/10.5281/zenodo.15733730> and <https://doi.org/10.5281/zenodo.15693025>. Pre-configured runtime environments including dependencies are provided via both a Conda environment (<https://anaconda.org/heritabilities/puzzler>) and an APTainer hosted both on Zenodo and Sylabs (<https://cloud.sylabs.io/library/merondun/default/puzzler>).

1 Introduction

High-quality, chromosome-level genome assemblies are instrumental across many areas of genomics. Accurate and contiguous assemblies are essential for investigating structural variation, evolutionary dynamics, and gene regulation (Rhie *et al.* 2021, Wright *et al.* 2024, Yoo *et al.* 2025). Moreover, pan-genomic approaches require dozens to hundreds of chromosome-scale assemblies (Liao *et al.* 2023, Secomandi *et al.* 2025). Advances in long-read sequencing, particularly PacBio high-fidelity (HiFi) reads with high accuracy, now enable construction of near-error-free gapless contigs. In parallel, Hi-C sequencing provides chromosome-scale contact information that facilitates scaffolding into chromosome-length assemblies. Together, these technologies can yield telomere-to-telomere assemblies and even phase haplotypes in diploid or polyploid organisms. Despite these advances, scalable and reproducible assembly remains a bottleneck. Some tools provide end-to-end tools for assembly, scaffolding, and quality control (Almeida *et al.* 2023, Larivière *et al.*

2024), while others address only isolated steps within the assembly workflow (Driguez *et al.* 2021, Angelova *et al.* 2022).

At present, building chromosome-scale genome assemblies typically requires pairing a long-read assembler with one or more Hi-C-based scaffolders (Rhie *et al.* 2021). Additional steps often include synteny-based labeling of chromosomes for comparative and pan-genomic applications (Liu *et al.* 2018). For small labs or users new to genome assembly, implementing best-practice workflows remains challenging, especially given that the final genome assembly is merely the starting point for downstream analyses (Jung *et al.* 2020). Scalability and reproducibility further compound the difficulty. Pan-genomic efforts benefit from standardized genome processing, necessitating pipelines that are both computationally efficient and easily portable across environments (Groza *et al.* 2024, Heumos *et al.* 2024). Existing workflows often rely on complex computing environments or manual curation, which hinder reproducibility and complicate scaling (Almeida *et al.* 2023, Larivière *et al.* 2024).

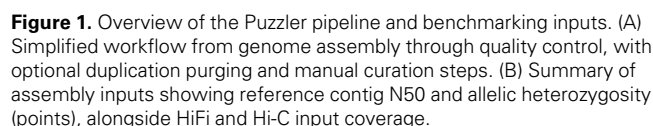
2 Methods and implementation

et al. 2020); purging can be skipped with `--no_purge`. The resulting contigs are then scaffolded into chromosome-scale structures using HapHiC v1.0.6 (Zhang *et al.* 2019, Zeng *et al.* 2024), which leverages chromatin contact data from Hi-C libraries aligned with bwa-mem2 v2.21, filtered using samblaster v0.1.26 and samtools v1.21 (Faust and Hall 2014, Vasimuddin *et al.* 2019, Danecek *et al.* 2021). A priori expected chromosome number is used as a target for scaffolding. If HapHiC fails to produce an assembly within ± 4 chromosomes of this target, the pipeline automatically invokes YAHF v1.2.2 as a fallback scaffolder (Zhou *et al.* 2023).

To assign chromosome labels, MashMap v3.1.3 (Kille *et al.* 2023) aligns scaffolds against the related species reference assembly, which is computationally efficient even for giga-base sized chromosomes. Alignments are parsed to assign chromosome names, assigning unlocalized fragmented scaffolds into labels such as “Chr1_unloc1”. Scaffold orientation is corrected by reverse-complementing scaffolds as needed with seqkit v2.10.0 (Shen *et al.* 2024). This reference-based chromosome naming has been internally validated across species diverged over 50 million years, including those with whole-genome duplications. While not strictly necessary for genome assembly, providing a chromosome-level reference with standardized chromosome labels improves interpretability of downstream analyses. This is especially useful in pan-genomic studies where consistent chromosomal labeling prevents misassignments and comparison errors.

Puzzler then proceeds to final quality control, including: (i) Hi-C contact map visualizations with HapHiC v1.0.6 (Zeng *et al.* 2024); (ii) k-mer-v0.1-r69 (Cheng *et al.* 2021); (iii) gene completeness via BUSCO v5.8.3 (Simão *et al.* 2015); and (iv) contamination screening with BlobTools v1.1.1 (Laetsch and Blaxter 2017) using blastn v2.16.0+ (Camacho *et al.* 2009) against NCBI's nt RefSeq database. Assembly metrics such as contig and scaffold N50 are calculated with assembly_stats v0.1.2 (Trizna 2020), and a final summary report is generated providing key statistics on genome size, contiguity, completeness, and potential contamination. Because Puzzler uses a sample sheet input and is check-point based, assembly can be executed at scale, facilitating high-throughput genome assembly.

We benchmarked Puzzler against 12 publicly available chromosome-scale eukaryotic genome assemblies generated from PacBio HiFi and Hi-C data, encompassing a wide taxonomic and genomic size spectrum (Table 1, available as supplementary data at *Bioinformatics Advances* online). All tests



were conducted across three distinct HPC environments using Puzzler v1.9, tested using both Sylabs containers and Conda environments. Test species included a ~24 Mbp fungal genome, a ~533 Mbp plant genome, an ~821 Mbp insect genome, and large vertebrate genomes, including a ~3.4 Gbp whale and a ~6.5 Gbp amphibian. All comparisons used assemblies constructed solely from HiFi and Hi-C reads, for which we used identical input files for Puzzler genome assembly (Table 2, available as supplementary data at *Bioinformatics Advances* online). Assemblies encompassed a range of HiFi input coverage (mean = 59.5×, range = 22.2–183.0×), mean HiFi read lengths (mean = 12.8 Kbp, range = 6.9–23.9 Kbp), and Hi-C input coverage (mean = 139×, range = 34.5–540.0×) (Fig. 1B). Only weak associations were observed between read inputs and assembly contiguity (maximum Pearson correlation = 0.43, Fig. 2, available as supplementary data at *Bioinformatics Advances* online).

Using 64 cores and 512 GB RAM, Puzzler demonstrated high computational efficiency across diverse genome sizes, with a mean assembly throughput of 1.8 Mbp/min (range = 0.5–3.8) (Table 3, available as supplementary data at *Bioinformatics Advances* online). Excluding manual curation and BlobTools, runtimes spanned from 32 min for the fungal genome (24.8 Mbp), to 3 h and 38 min for the fly genome (817 Mbp), and 44 h and 19 min for an amphibian genome (6478 Mbp). Resource monitoring showed a mean peak CPU usage of 39 cores (range = 13–47) and mean RAM usage of 336 GB (range = 58–512), indicating diminishing returns above 48 CPUs (Table 4, available as supplementary data at *Bioinformatics Advances* online). Notably, all genomes smaller than 1.4 Gbp were successfully assembled using 16 CPUs and 128 GB of RAM, although this increased assembly time by nearly 1.5-fold (mean = 1.5×, range = 0.2–2.4×; Table 5, available as supplementary data at *Bioinformatics Advances* online). BlobTools substantially increases runtimes; fly genome (821 Mbp) screening with 48 cores against NCBI nt took an extra 24 h.

Puzzler generated high-quality, chromosome-scale assemblies across diverse taxa and genome sizes as evidenced by clear chromosome structures in the Hi-C contact maps (Fig. 1, available as supplementary data at *Bioinformatics Advances* online). For the 12 test species, assemblies

generated with Puzzler—incorporating a manual curation step—exhibited a median contig N50 of 12.6 Mbp (range = 1.4–68.7 Mbp) and a scaffold N50 of 72.7 Mbp (range = 1.8–743.5 Mbp) (Fig. 2 and Table 6, available as supplementary data at *Bioinformatics Advances* online). The median BUSCO completeness score across all Puzzler assemblies was 98.6% (range = 93.8–99.5%), slightly outperforming their corresponding published references (mean Δ = +0.2%, range = −0.3%–1.8%; Fig. 2 and Table 7, available as supplementary data at *Bioinformatics Advances* online). Similarly, scaffold N50 (mean Δ % = +5.2%, range = −1.1%–27.1%) was higher in Puzzler assemblies, although they also contained more sequences (150.9%, range = −78.3%–1361.9%). Overall, assembly sizes showed only minor deviations from the reference, with a median percent change of just 1.4% (range = −7.7%–8.3%; Fig. 2 and Table 8, available as supplementary data at *Bioinformatics Advances* online). We note that case-specific attention is needed for duplicate purging, as this step showed highly variable outcomes across the 12 assemblies tested (Fig. 4, available as supplementary data at *Bioinformatics Advances* online).

Overall, performance was consistent across genomes of varying size and complexity, from a ~24 Mbp fungal genome to a ~6.5 Gbp amphibian genome, demonstrating that Puzzler scales effectively across a broad range of chromosomal structures. This holds across genomes with heterozygosity ranging from haploid *Rhizosphaera kalkhoffii* to over 2% in *Ascidia mentula*, highlighting Puzzler's ability to handle both low-diversity and highly heterozygous assemblies (Table 9, available as supplementary data at *Bioinformatics Advances* online). Whole-genome alignments confirmed highly syntenic assemblies between Puzzler and published references, even when run in fully automated, uncured mode using—no_juice (Fig. 3). Across all samples, Puzzler yielded 33 fully gapless chromosomes out of 220 total (15.0%), compared to 43 (19.5%) in the corresponding reference assemblies (Fig. 5, available as supplementary data at *Bioinformatics Advances* online). Notably, only three chromosomes were absent from the Puzzler assemblies (chromosomes 30–32 in the Eurasian Crane, lengths 1.7–2.1 Mbp). These chromosomes were recovered in the—no_juice assemblies without manual curation but appear to have been

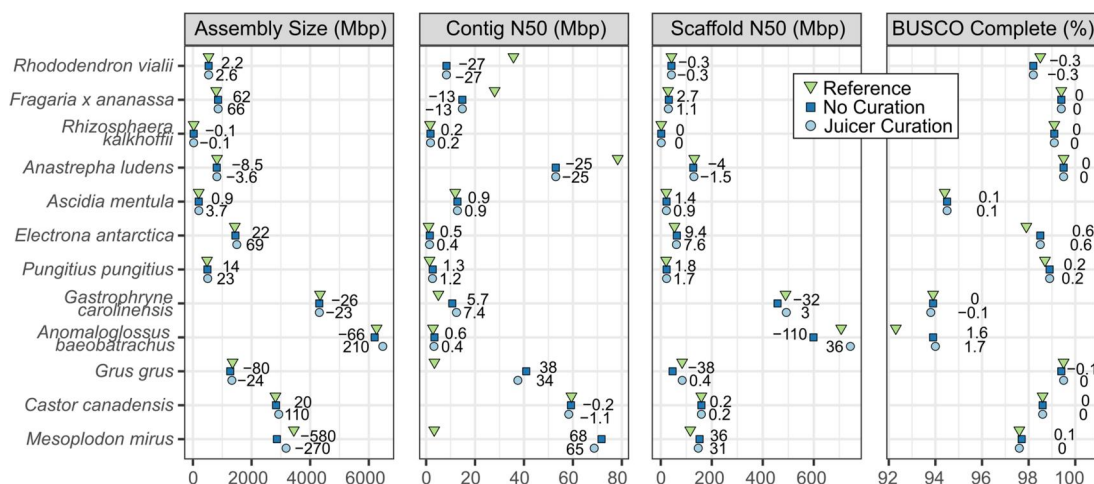


Figure 2. Comparative metrics for Puzzler versus reference assemblies, with deltas showing relative differences (positive = higher in Puzzler, negative = higher in reference). Statistics for Puzzler assemblies are shown with and without manual curation.

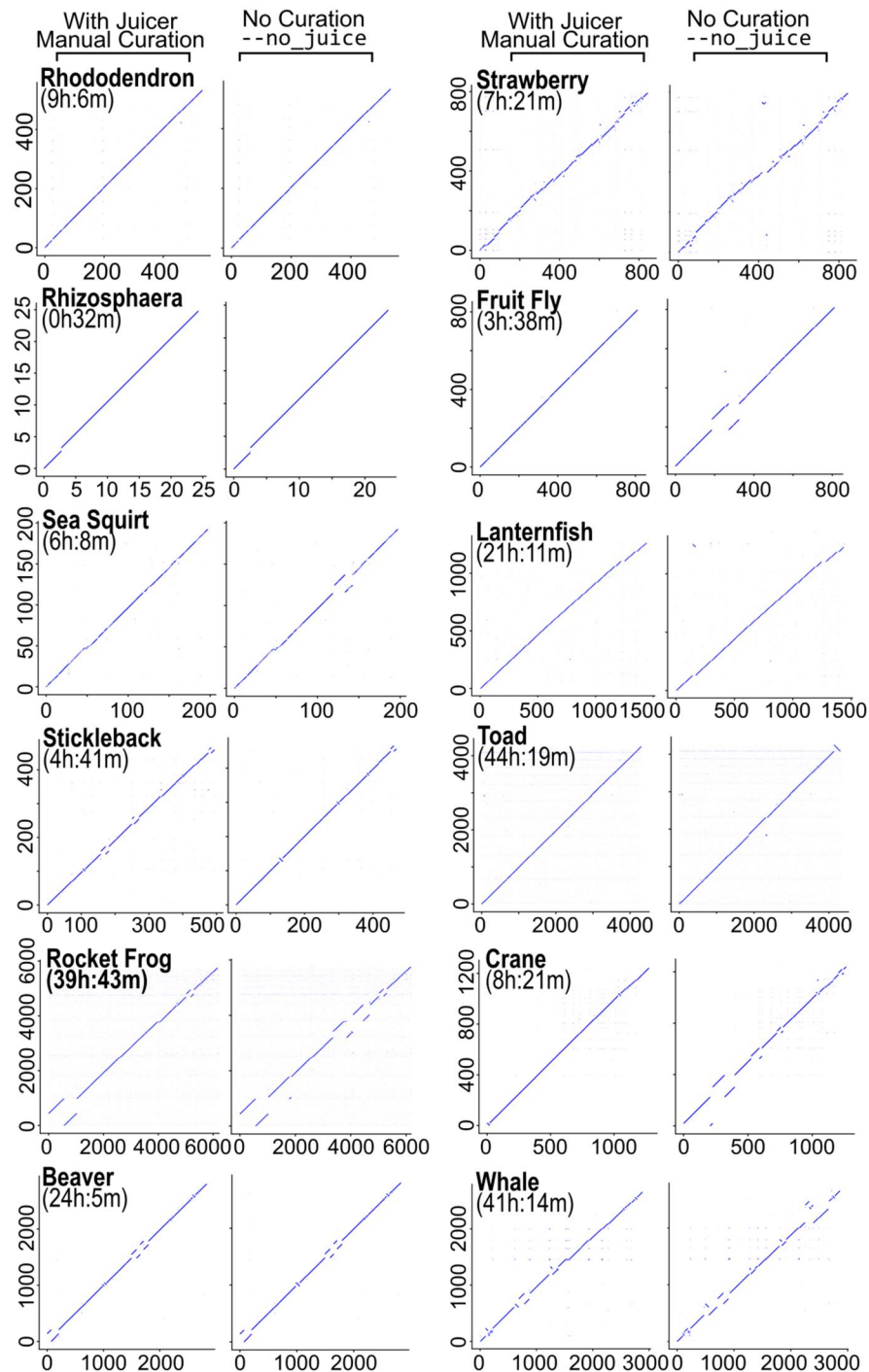


Figure 3. Whole-genome alignments between reference genomes and Puzzler draft assemblies generated with and without manual curation. Dotplots show synteny (reference on y-axis, Puzzler on x-axis; Mbp scale), assembly times reported for 64 cores and 512 GB RAM.

misjoined during juicer curation. This illustrates the complexity of manual curation (Fig. 3, available as supplementary data at *Bioinformatics Advances* online), where detailed scrutiny and iterative validation are essential to ensure that small, structurally complex chromosomes are correctly resolved.

4 Conclusion

Our findings demonstrate that automated workflows are capable of matching, if not surpassing, the structural standards established by platinum-grade reference assemblies curated

by expert bioinformatics teams. As chromosome-level assemblies become increasingly central to functional, ecological, and evolutionary genomics, Puzzler provides a scalable and reproducible solution for generating high-contiguity assemblies across large sample sets.

Puzzler is a containerized, end-to-end pipeline for *de novo* genome assembly and Hi-C scaffolding using long-read data. It automates best-practice workflows with a single command, combining contig assembly, scaffolding, and quality control into a reproducible and user-friendly framework. Designed for scalability and reproducibility, Puzzler supports genome

sizes from fungi to frogs, generates Juicebox-compatible outputs for manual curation, and embeds checkpoints to prevent re-computation, facilitating integration into large assembly projects and pan-genomic efforts.

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Supplementary material

[Supplementary material](#) is available at *Bioinformatics Advances* online.

Conflicts of interest

None declared.

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Data availability

Videos detailing manual curation available at: <https://www.youtube.com/watch?v=rMUINqZwEpA>.

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